



Quadratic graphs

THE GRAPH OF A QUADRATIC EQUATION IS A SMOOTH CURVE.

The exact shape of the curve of a quadratic graph varies, depending on the values of the numbers a , b , and c in the quadratic equation $y = ax^2 + bx + c$.

Quadratic equations all have the same general form: $y = ax^2 + bx + c$. With a particular quadratic equation, the values of a , b , and c are known, and corresponding sets of values for x and y can be worked out and put in a table. These values of x and y are then plotted as points (x, y) on a graph. The points are then joined by a smooth line to create the graph of the equation.

A quadratic equation can be shown as a graph. Pairs of x and y values are needed to plot the graph. In quadratic equations, the y values are given in terms of x —in this example each y value is equal to the value of x squared (x multiplied by itself), added to 3 times x , added to 2.

$$y = x^2 + 3x + 2$$

y value gives position of each point on y axis of the graph

this group of terms is used to find the y value for each value of x

Find sets of values for x and y in order to plot the graph. First, choose a set of x values. Then, for each x value, work out the different values (x^2 , $3x$, 2) for each value at each stage of the equation. Finally, add the stages to find the corresponding y value for each x value.

choose some values of x around 0

$y = x^2 + 3x + 2$, so it is difficult to work out the y values right away

work out x^2 in this column

work out $3x$ in this column

+2 is the same for each x value

add values in each purple row to find y values

x	x^2	$3x$	+2	y
-3	9	-9	2	
-2	4	-6	2	
-1	1	-3	2	
0	0	0	2	
1	1	3	2	
2	4	6	2	
3	9	9	2	

x	x^2	$3x$	+2	y
-3	9	-9	2	2
-2	4	-6	2	0
-1	1	-3	2	0
0	0	0	2	2
1	1	3	2	6
2	4	6	2	12
3	9	9	2	20

y is the sum of numbers in each purple row

△ Values of x

The value of y depends on the value of x , so choose a set of x values and then find the corresponding values of y . Choose x values either side of 0 as they are easiest to work with.

△ Different parts of the equation

Each quadratic equation has 3 different parts—a squared x value, a multiplied x value, and an ordinary number. Work out the different values of each part of the equation for each value of x , being careful to pay attention to when the numbers are positive or negative.

△ Corresponding values of y

Add the three parts of the equation together to find the corresponding values of y for each x value, making sure to pay attention to when the different parts of the equation are positive or negative.

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◀ 192–193 The quadratic formula